Elvis Aguero

PhD. Candidate in Engineering. MSc. Applied Mathematics

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Education

2024-2029 **PhD, Fluids and Thermal Sciences**, *Brown University*, Providence, RI. **Brown Diversity Fellow**.
Advancing multidisciplinary research that bridges applied mathematics, scientific computing, and data-driven modeling of complex fluid systems.

2020-2024 **B.Sc., Engineering Physics**, *Universidade Federal da integração Latino-Americana (UNILA)*, Foz do Iguaçu, Brazil. **Summa cum Laude**.

2017–2019 **M. Sc., Applied Maths**, *Instituto de Matemática Pura e Aplicada*, Rio de Janeiro, Brazil.
Completed Master's studies at 18. Advisor: André Nachbin

Publications & Technical Work

2025 E. A. Agüero, C. A. Galeano-Rios, C. T. Gabbard, C. Ragazzo, D. M. Harris, and P. A. Milewski. *Droplet rebounds off a fluid bath: kinematic match simulations and experiments*. **Arxiv preprint under review in the Journal of Fluid Mechanics**. arXiv:2509.22826v1 Led numerical modeling, implementation and analysis.

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2025 C. T. Gabbard, E. A. Agüero, R. Cimpeanu, K. Kuehr, E. Silver, J.-W. Barotta, C. A. Galeano-Rios, and D. M. Harris. *Drop rebound at low Weber number*. **Journal of Fluid Mechanics**, vol. 1019, A25. DOI:10.1017/jfm.2025.10589. Led numerical modeling, implementation and analysis.

2022 E. A. Agüero, L. Alventosa, D. M. Harris, and C. A. Galeano-Rios. *Impact of a rigid sphere onto an elastic membrane*. **Proceedings of the Royal Society A**. DOI:10.1098/rspa.2022.0340. Led numerical modeling, implementation and analysis.

Experience

2017-2024 **Olympiad Organizer and Trainer**, *OMAPA*, Asunción, Paraguay (Remote).
- Author of more than 200 math problems for the Paraguayan National Math Olympiad, as well as international math competitions.
- Instructor for the Young Talents Program, where the best students from the National Olympiad receive mathematical and scientific training for international competitions.

2024 **Summer Intern**, *CERN*, Geneva, Switzerland.
- Worked in the department of software for experimental physics at CERN (EP-SFT), where we used Julia to leverage open-source, large-scale workflows to enable postprocessing to run at scale.

2023 **Scanning Electron Microscope Lab Assistant**, *UNILA*, Foz do Iguaçu, Brazil.
- Provided administrative assistance and prepared samples for material characterization for the Scanning Electron Microscope under the guidance of Prof. Rodrigo Basso

2020-2023 **Academic Consultant**, *Piensa Paraguay*, Ciudad del Este, Paraguay.
- Designed the math curriculum and devised the structure and maintenance of a database with more than 1000 math problems in MongoDB. Piensa is a local startup which offers math, robotics and programming courses for students aged $7-17$.

2019 **Project Intern**, *Parque Tecnológico Itaipu*, Hernandarias, Paraguay.
- Implemented a support vector machine algorithm in MATLAB for classification of faulty high-voltage circuit breakers.

Fellowships and Awards

2026 **Hazeltine Innovation Award**, Brown School of Engineering. Co-investigator on *"Autonomous Discovery of Bioreactor Designs via Agentic Multi-Fidelity Bayesian Optimization"*. Link

2024-2026 Diversity Fellow at Brown University.

2017 CNPQ Fellowship for M.Sc. Studies at IMPA.

2017 **Prize "Domingo Martinez de Irala"**: Order of merit Medal, Asunción, Paraguay.

2015, 2016 **International Math (IMO)**: Bronze Medals (2015, 2016).

2015, 2016 **IberoAmerican Math Olympiad**: Honorific Mention (2015), Bronze Medal (2016).

2011-2016 **Paraguayan Math Olympiad**: Gold Medals (2013, 2015, 2016) and Silver Medals (2011, 2012, 2014).

2016 **Distinguished Citizen of Ciudad del Este Medal**.

Talks

2025 Nov *Wave-driven propulsion of a flexible raft*. Contributed talk, **APS DFD (78th Annual Meeting)**, Houston,Texas, USA.

2025 Oct *The kinematic matching method for deformable impactors*. **Numerical Methods Seminar**, Instituto de Matemática e Estatística, Universidade de São Paulo, Online presentation.

2024 Nov *Rebound of a droplet impacting a non-wetting rigid surface*. Contributed talk, **APS DFD (77th Annual Meeting)**, Salt Lake City, Utah, USA. Link

2024 Aug *Enabling Julia to run at Scale with artifact caching*. Contributed talk, **JuliaHEP Workshop 2024**. Online presentation, held at CERN, Switzerland. Link

2023 Jul *Impact of a rigid sphere onto an elastic membrane*. Contributed talk, **Brazil–China Joint Mathematical Meeting**, Foz do Iguaçu, Brazil. Program

2021 Dec *Modelado de impacto axisimétrico de un sólido sobre una membrana elástica*. Talk, **4to Ciclo de Charlas Científicas "Creando Redes"**, online, Paraguay. Video

2021 Nov *Modelado de impacto axisimétrico de un sólido sobre una membrana elástica*. Talk, **CPMyCA — Coloquio Paraguayo de Matemática y Ciencias Afines**, Paraguay.

Teaching

Fall 2025 Teaching Assistant, **Data-Driven Design & Analysis of Structures & Materials**, Brown University. Course Website

Summer 2025 Teaching Assistant, **Soft Matter Physics**, Brown University Pre-College Program.

2017-2024 Instructor, Young Talents Program, OMAPA. Videos.

Fall 2022 Teaching Assistant, **Physics II**, UNILA, Foz do Iguaçu, Brazil.

Fall 2021 Teaching Assistant, **Numerical Calculus**, UNILA, Foz do Iguaçu, Brazil.

Summer 2019 Teaching Assistant, **Functional Analysis (Summer School)**, IMPA, Rio de Janeiro, Brazil.

Workshops and Summer Schools

2025 Participant, **Summer School on Soft Matter Physics**, UMass Amherst, Amherst, MA. Topics included: Control Theory in Physics, Physics of Biofilms, Instability, Flocking and Self-assembly.

2024 Participant, **7th Summer School in Advanced Experimental Physics**, CBPF, Rio de Janeiro, Brazil (acceptance rate $\approx 15\%$). Hands-on modules included ferromagnetic resonance, magnetron sputtering, and X-ray reflectivity to study thin-film bilayers.

Programming & Technical Skills

Experienced

- Julia: Scientific Computing, OOP.
- MATLAB: Scientific Computing, OOP, Parallel computing, GPU-accelerated algorithms.
- Python: Scientific Computing, OOP, and ML programming with Pytorch/JAX.
- Git, Github, bash: Remote and local version control, contributed to a number of open-source projects.

Familiar

- C++: OOP, Scientific Computing.
- Java, JS, Google Apps Script, SQL, MongoDB, Google Sheets, Access, LabView, R, HTML, Scilab.

Volunteer Work and Outreach

2017-Present **Math Olympiads Coordinator, Jury member, Team Leader**, *OMAPA*.

- Part of the Paraguayan delegation as organizer in numerous International Olympiads, such as the Cyberspace Mathematical Competition (2020), IberoAmerican Olympiad (2021), and the International Mathematical Olympiad (2022), which was held in Oslo, Norway.
- Part of the jury committee at the XXXII Cono Sur Olympiad (2021), organized by Paraguay. Graded problems of all countries, and created problems for the contest.

2022-2023 **Engineering Physics Student Council**, *UNILA*.

2019-2022 **Board member, Teaching Staff**, *Mathura*.

- Board member of Mathura, an NGO which gives free Maths and Spanish classes to underserved high school students in their pursuit to undergraduate studies. Also, part of the math teaching staff of Mathura.

2019-2021 **Volunteer Data Analyst**, *ReAcción Paraguay*.

- ReAcción is a non-profit NGO that uses civic education and grassroots mobilization to demand accountability from government in the education sector.
- Used Python and Apps Script to analyze thousand of public contracts, comparing public expenditure of central and local institutions before and after the covid-19 outbreak. A couple of articles (in Spanish) were made based on our findings: `https://reaccion.org.py/publicaciones/#investigaciones`
- Gave seminars to the public to raise awareness on the importance of open data, and worked on a project to automate the process of contract data analysis in our country.

Languages

Spanish	CEFR C2	Portuguese	CEFR C2
English	CEFR C1	French	CEFR A1

Other Activities

- Problem solving: My ProjectEuler account is percentile 0.4% among all problem solvers.
- When I'm not solving problems, I am usually playing some chess on Lichess!.
- When I'm not playing chess, I am usually running or playing football.